

MAANAS RASTOGI

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EDUCATION

Bachelor of Science, Electrical/Computer Engineering

Integrated Circuits and Systems

University of Texas at Austin

Expected May 2028

GPA: 3.7/4.0

PROFESSIONAL EXPERIENCE

Validation/Design Intern | Ironlattice

August 2026 - Present

- Developing simulation pipelines using COMSOL, atomistic tools (QuantumATK), and Synopsys TCAD to model device physics, gate-stack parameters, and array-level SPICE behavior for the 40 nm node roadmap.
- Modeling Gator Memory™ 1-FeFET superlattice architectures $4F^2$, analyzing ferroelectric switching trade-offs targeting 10^{10} cycle endurance, 10 ns speed, and 30+ analog states for compute-in-memory.

Racing Team Electronics Lead | Longhorn Baja Racing

August 2025 - Present

- Direct vehicle electrical integration for the Baja SAE race car, designing 20-gauge power distribution, custom dashboard hardware, and a virtual driving coach interfacing with 7+ onboard sensors.
- Leading a team of 10+ members, organizing technical workshops on Altium Designer, circuit theory, and PCB design while driving design reviews and race-day telemetry validation.
- Developing control algorithms to translate 3D spatial mapping into automated steering and throttle inputs with less than 50 ms processing delay.

Quantum Computing Undergraduate Researcher | The University of Texas at Austin

October 2025 – December 2025

- Engineered a simulation pipeline utilizing Stim, sinter, and PyMatching to model surface-code stabilizer circuits across modular, chiplet-based quantum processors.
- Executed 100,000+ Monte Carlo simulations to quantify logical error scaling, specifically analyzing the impact of 1% inter-chip network noise on fault tolerance.

Manufacturing/Design Lead | Longhorn Neurotech

August 2024 - August 2026

- Directed a team of 5 engineers to develop modular electronics mounting systems for a 250-lb electric-powered wheelchair, reducing hardware integration time by 30%.
- Implemented motor control for a drone, using a potentiometer and wireless signals to regulate motor speed, improving responsiveness and flight stability by precisely mapping analog input to motor output with a latency of 10 milliseconds.

Electrical Engineer | UT Austin Robotics & Automation Society

August 2024 - August 2025

- Improved power delivery and efficiency of a robotics system, boosting total output from 12W to 90W by redesigning the supercapacitor bank and implementing a new current-limiting circuit, increasing the system's runtime by 20%.
- Redesigned the main control board to improve accessibility and maintenance, reducing system charging times by 75% by increasing the charging current to 6A via a high-efficiency charging circuit.

TECHNICAL EXPERIENCE

VLSI Design

- Designing a 32-bit custom RISC-V processor to drive a full-stack ASIC development flow from architectural specification to physical layout.
- Executing HDL programming, simulation, and synthesis utilizing Verilog within the Vivado environment.
- Hardening digital designs into physical silicon via OpenROAD, focusing on optimization constraints to achieve a target clock frequency of 50 MHz and reduce area overhead by 15%.

Motorcycle Helmet HUD

- Built a Raspberry Pi Android Auto head unit with a 5-inch touchscreen, achieving sub-100 ms input latency for a responsive UI.
- Programming Python controls and designing a prism-based optical display projecting 60 Hz ride data across a 20-degree FOV with 85% visor transparency.
- Engineering a 5V / 3A power network with a 3000 mAh battery, delivering 4+ hours of runtime for embedded components within a 50 cm³ profile.

Gamble on the Go

- Engineered a portable blackjack game for the TI LP-MSPM0G3507 board, creating a user-selectable English and Spanish interface to improve accessibility and user experience, and integrated sound design via an ADC
- Developed a hardware UI utilizing 4 tactile buttons and a 10k-ohm slide potentiometer, winning 1st place out of 15+ projects in the class competition.

SKILLS

Programming: Python, Java, Assembly, C/C++, Verilog

Development: Circuit Board Design, Microcontroller Operation, Quantum Computation, Simulation

Soft Skills: Communication, Leadership, Collaboration, Time Management, Adaptability